

Good practice R coding

Day 1 - Introduction to Data Analysis with R

Selina Baldauf

Freie Universität Berlin - Theoretical Ecology

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Chaotic projects and workflows ...

... can make even small changes frustrating and difficult.



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Set up your project properly

Consistent structure and filenames

Have a clear project structure

- One directory with all files relevant for project
 - Scripts, data, plots, documents, ...
- Choose a meaningful project structure ¹
- Keep your raw data untouched
 - `data/raw` for the original data
 - `data/clean` for cleaned data created by your scripts
- Add a readme file (usually `README.md`) in which you document the project structure

```
MyProject/  
├── data/  
│   ├── raw/  
│   └── clean/  
├── R/  
├── figures/  
├── MyProject.Rproj  
└── README.md
```

Example RStudio project structure

1. you can orient yourself at the [R package structure](#)

Use informative file names

- Human readable: reveal the content
- Machine readable: no spaces, no special characters
- Work with default ordering: left-padded numbers or YYYY-MM-DD dates first

```
MyProject/  
├── data/  
│   ├── raw/  
│   │   └── 2026-10-12_growth-assay.csv  
│   └── clean/  
│       └── growth-assay_clean.csv  
├── R/  
│   ├── 01_clean-data.R  
│   ├── 02_fit-models.R  
│   └── 03_make-figures.R  
└── figures/  
    └── linear-model-fit.png
```

Let's start coding

Good practice R coding

Write beautiful code

- Try to write code that others (i.e. future you) can understand
- Follow standards for readable and maintainable code
 - For R: [tidyverse style guide](#) defines code organization, syntax standards, ...



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Standard code structure

1. General comment with purpose of the script, author, ...
2. `library()` calls on top
3. Set default variables and global options
4. Write the actual code, starting with loading all data files

```
# Analysis of tree growth data
# Author: Selina Baldauf
# Date: 2026-03-09

library(tidyverse)

# set defaults
input_file <- "data/tree_growth.csv"

# read input
trees <- read_csv(input_file)

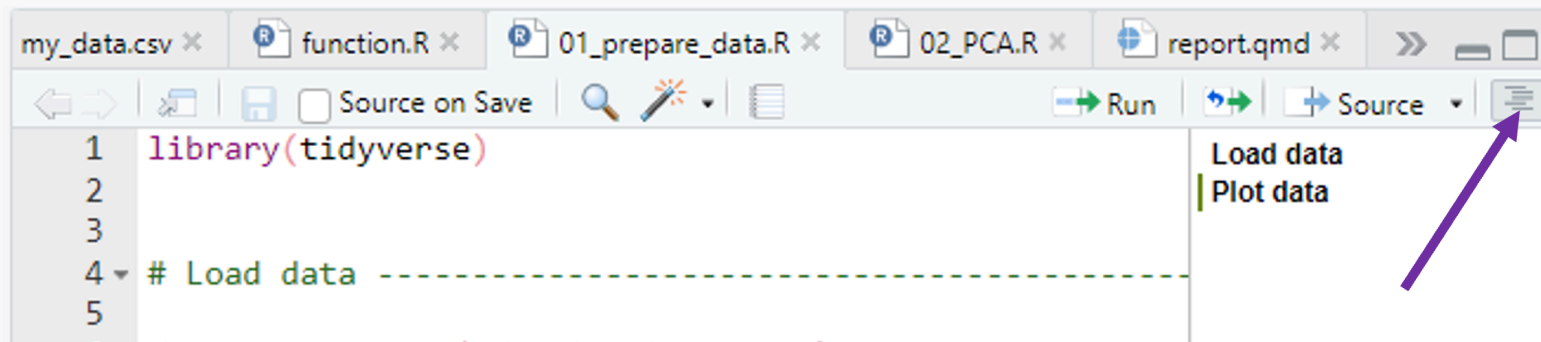
# analyse data
summary(trees)
```

Mark sections

- Use comments to break up your file into sections

```
# Load data -----  
  
trees <- read_csv("data/tree_growth.csv")  
  
# Explore data -----  
  
summary(trees)
```

- Insert a section label with **Ctrl/Cmd + Shift + R**
- Navigate sections in the file outline



Coding style - Object names

- Variables should only have *lowercase letters, numbers, and *_**
- Use **snake_case** for longer variable names
- Try to use concise but meaningful names

```
# Good
day_one
day_1

# Bad
DayOne
dayone
first_day_of_the_month
dm1
```

Coding style - Spacing

Use spaces to make your code more readable:

- Always put spaces after a comma
- Spaces around most operators (<-, +, etc.)
- No spaces around parentheses for normal function calls

```
# Good
x[, 1]
height <- (feet * 12) + inches
mean(x, na.rm = TRUE)
```

```
# Bad
x[,1]
height<-feet*12+inches
mean(x,na.rm=TRUE)
```

Coding style - Line width

Try to limit your line width to 80 characters.

- You don't want to scroll to the right to read all code
- 80 characters can be displayed on most displays and programs
- Split your code into multiple lines if it is too long
 - See this grey vertical line in R Studio?

Coding style

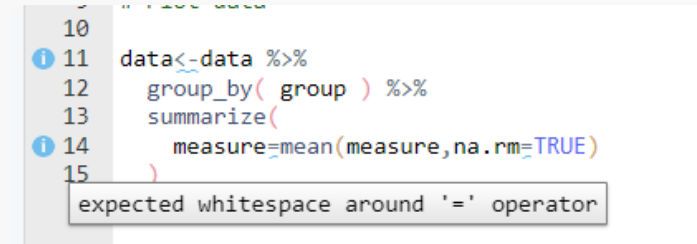
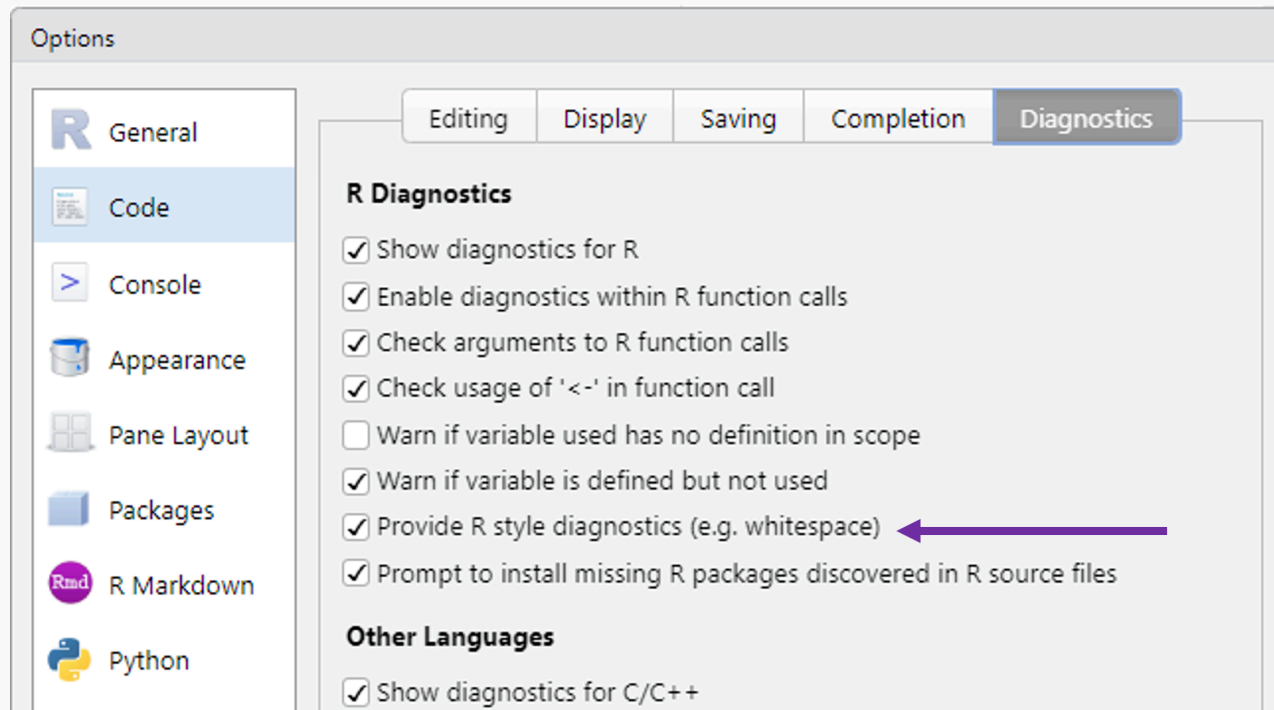
Do I really have to remember all of this?

Luckily, no! R and R Studio provide some nice helpers

Coding style helpers - RStudio

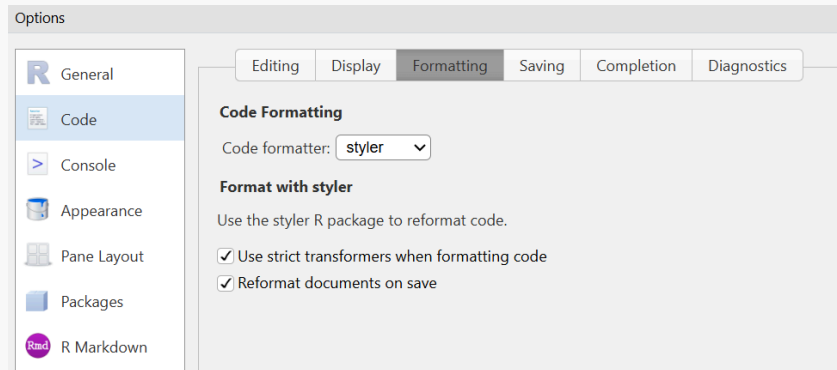
RStudio has style diagnostics that tell you where something is wrong

Tools -> Global Options -> Code -> Diagnostics



Coding style helpers - Auto-formatting

RStudio can automatically format your code!



Try it now: open your script from today and use **Code -> Reformat Code** to auto-format it.

Clean projects and workflows ...

... allow you and others to work productively.

But don't get overwhelmed by all the advice. Just start with one thing.



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